# Synthetic Instance Configuration 50

## Generated Files

| Instance file | $\|S\|$ | $\|P\|$ | $\|B\|$ | $\|E\|$ | $\|T\|$ | $\|I\|$ | Seed |
|---|---|---|---|---|---|---|---|
| problem_instance_ext_50_0.txt | 50 | 60 | 8 | 5 | 15 | 8 | 1246 |
| problem_instance_ext_50_1.txt | 50 | 60 | 7 | 5 | 12 | 7 | 1247 |
| problem_instance_ext_50_2.txt | 50 | 60 | 6 | 4 | 8 | 6 | 1248 |
| problem_instance_ext_50_3.txt | 50 | 60 | 9 | 4 | 11 | 3 | 1249 |

$|S|$: students; $|P|$: placements; $|B|$: companies; $|E|$: educational centres; $|T|$: professors; $|I|$: dependency groups; Seed: seed used to generate the instance.

## Generator Arguments

| Argument | Description | Value |
|---|---|---|
| result_path | Output folder where the generated files are written | output |
| start_size | First number of students considered | 50 |
| end_size | Last number of students considered | 50 |
| number_instances | Number of instances generated for each student size | 4 |
| surplus | Extra placements percentage with respect to students | 20 |
| slack_ratio | Extra capacity margin for company and professor quotas | 0.2 |
| placement_dep_prob | Percentage used to create placement dependency groups | 30 |
| number_competences | Number of base competences per profile | 6 |
| most_important_competences | Number of primary competences required by a placement | 2 |
| seed | Base seed used to derive the seed of each generated instance | 1246 |

## Internal Parameters

| Parameter | Description | Value |
|---|---|---|
| student_size_step | Fixed step between generated student sizes | 200 |
| number_profiles | Number of academic profile blocks | 8 |
| p_excellent_student | Probability of generating an excellent student | 0.05 |
| p_notable_student | Probability of generating a notable student | 0.25 |
| car_probability | Probability that a student has access to a car | 0.65 |
| student_distance_weight_range | Range used for the student distance weight | $[0.0, 0.5]$ |
| n_centers_min | Minimum number of educational centres | 2 |
| n_centers_max | Maximum number of educational centres | 6 |
| prof_capacity_min | Minimum professor supervision capacity | 5 |
| prof_capacity_max | Maximum professor supervision capacity | 25 |
| ORS_AVAILABLE | Whether ORS routing support is available in the script | False |
| use_ors | Whether ORS is actually used for this generation | False |
| ors_base_url | ORS endpoint used when routing is enabled | http://localhost:8080/ors |
| synthetic_duration_mu | Mean of the synthetic duration distribution | 1500 |
| synthetic_duration_sigma | Standard deviation of the synthetic duration distribution | 250 |

## Randomness

| Field | Description | Value |
|---|---|---|
| Base seed | Initial seed provided to the generator | 1246 |
| Seed policy | Rule used to derive instance seeds from the base seed | instance_seed = base_seed + global_instance_idx |
| Python random seeded | Whether the Python random generator is seeded per instance | Yes |
| NumPy random seeded | Whether the NumPy random generator is seeded per instance | Yes |